

7.2.3 Main Sources of Uncertainty in the Laboratory

To fully explain how this approach can be useful to quickly estimate the MU, it is necessary to review five major causes of Figure 7.2.

7.2.3.1 Manpower

This is a delicate point since the aim is to assess the *competence* of the personnel responsible for the analyses. The ISO 17025 standard provides for the organization of training programs for the personnel, a qualification procedure for any new person entering the laboratory, and another for maintaining this competence. It is obvious that the implementation of a school-type examination, such as the analysis of material of known content, cannot be considered sufficient in a random context where acceptable deviations are not yet defined.

Sources of variability commonly attributed to the operators include systematic bias in the reading of a signal, weighting or volume error, and a different interpretation of a result, but all are correctable.

7.2.3.2 Material and Handling of Items

These sources of uncertainty can generally be considered to be part of the pre-analytical phase. But, when samples are stored for a certain period of time prior to the analysis, the storage conditions may affect the results. The storage time and conditions should therefore be considered part of the MU. If sampling is the responsibility of the laboratory, the uncertainty associated with this operation must be evaluated. This is still currently under discussion, as it is difficult to achieve.

Some elements are presented in Section 8.3, and others are available in [18]. As soon as sampling is mentioned, many situations have to be considered, each of them being specific and requiring a particular statistical formulation: sampling plants in a field, blood in medical biology laboratories, or samples on a bulk lot of raw materials, etc. are all different in terms of statistical modeling. In addition, heterogeneity within a batch creates random variations between different samples that are not always easy to model.

Finally, the sampling procedure itself can produce a bias. For example, a water sampling device in a river creates a vortex that may displace some suspended particles.

7.2.3.3 Method

This is probably where analysts best identify the sources of uncertainty. Most analytical procedures are a combination of operations, especially if the assay requires preparation. Without going into too much detail, it is necessary to sort out parameters that can globalize these sources of uncertainty. For example, the analyte recovery yield in a complex matrix can serve this purpose, even though the instrumental response may be affected by matrix effects. Analyte speciation can also increase these effects when a spike is used to estimate a recovery rate and the surrogate substance is in a chemical form different from the analyte.

Metrologists like to point out that often glassware material used at a different ambient temperature than that at which it was calibrated is a source of uncertainty. These effects are usually negligible, especially if the laboratory has air conditioners, essential for many modern measuring instruments. On the other hand, sample moisture can play a significant role if it is sensitive to possible hygrometric changes, which happens with biological samples that dry-freeze. There are also diverse sources of uncertainty due to calculations and signal processing. The selection of an approximate calibration model, early truncation and rounding of raw measurements, and incorrect settings of an integrator; all of these can lead to inaccuracies in the result.

The accuracy profile method can be used to detect these errors but not always to correct them. In trace analysis, blank or inherent concentration subtraction is significant. Uncertainty affects both the result and the timeliness of the blank correction. This problem is also encountered with the accuracy profile when standard additions are used, and it has not been possible to find a validation material without inherent content. Subtracting the so-called *no-addition* concentration also increases the MU (Chapter 1).

Finally, ISO 17025 cites an extreme situation where the analytical process is supposed to follow a certain stoichiometry, and the reaction proceeds in an incomplete way or with side reactions; then, it may be necessary to tolerate deviations from the expected stoichiometry.

7.2.3.4 Machine/Equipment

Measurement equipment affects the MU, for example, the limited accuracy of the laboratory scales, the difference between indicated settings and actual temperature for a thermostat, the effects of contamination or memory for an automatic injection device, and many other technical downsides. That is why the instruments of an accredited laboratory must be regularly qualified and subject to regular maintenance contracts and control charts, all monitoring designed to reduce the risks of malfunction or drift.

For reagents, the concentration of a calibration solution is not known exactly. Many commercially available chemical compounds are not 100% pure and may contain isomers and mineral salts. The purity of these substances is usually provided by the manufacturer as a lower limit. Any assumption about the degree of purity introduces an element of uncertainty.

7.2.3.5 Environment

The use of poorly designed spaces influences uncertainty. Some equipment requires air-conditioned rooms or clean rooms with positive pressure. The most striking example is that of microbiological methods, which require dedicated buildings known as *walk-in* rooms designed to avoid any cross-contamination in such a way that new samples never come in contact with already analyzed samples.

7.2.3.6 Measurement and Other Sources

First, any statistical model must consider that there are random, unlisted, or overlooked effects, generally referred to as residuals, which contribute to uncertainty.

They are inevitably included in the list. There remain two sources of uncertainty that deserve special mention:

- Traceability of measurements. This issue is dealt with exhaustively by the Bureau international des poids et mesures (BIPM).
- Sampling uncertainty needs special attention and is discussed in Section 8.3.

7.2.4 Stages 3 and 4. Calculation of Combined Uncertainty

Looking at the diagram of Figure 7.2 it should be possible to propose an experimental method based on the multifactorial analysis of variance (n -way ANOVA) to estimate the standard uncertainty of each MU component. The latter will extract the variance component of each source in an equivalent way as between- and within-variances calculated in Section 3.2.

This statistical procedure is the generalization of the one-factor ANOVA. But it requires a substantial experimental effort, i.e. a large number of runs, as well as professional statistical software capable of handling complex models that may combine fixed-effect and random-effect factors. Therefore, only a limited number of publications are available to illustrate this procedure [19, 20].

However, at stage 3 of the GUM general procedure, it is judicious to aggregate the standard deviations of several identified components before evaluating them, especially when numerous components are present or difficult to estimate. In Figure 7.2, the dotted area exhibits together six of the eight sources of uncertainty. It is labeled as a *series effect* because it combines the method of analysis and the practical conditions of its application when the analysis is carried out.

Indeed, the same sample is reanalyzed over several days or in any other way that represents the so-called intermediate precision condition. All the sources of uncertainty grouped in the “series effect” will be brought into play simultaneously. In Section 5.4.2, the need to use an experimental design establishing intermediate precision conditions was pointed out for constructing the accuracy profile. Thus, the simplest and most practical approach consists in calculating an accuracy profile that can be used to estimate MU. It remains to be seen whether the estimation of the standard uncertainties and, ultimately, the combined standard uncertainty can be easily derived from these data.

This demonstration requires some simple mathematical operations. It starts with the generic measurement model of Z_m described by Eq. (7.1), which is a simple additive polynomial relationship. After the law of propagation of uncertainty is applied, as shown in the Eqs. (6.11–6.13), the standard uncertainty of $u(Z_m)$ is given by the formula (7.4). For ease of calculation, the reference value X is considered constant and known without uncertainty, in other words, $u(X) = 0$. Moreover, the quantities $\sum_{n=1}^N c_n^2 \times u^2(G_n)$ are assumed to be negligible compared to the other sources of uncertainty. Because both assumptions are plausible, this leads to the simplified formula of the Eq. (7.6).

Measurement model

$$Z_m = X + \delta + B + \sum_{n=1}^N c_n G_n + E \quad (7.1)$$

Standard uncertainty

$$u(Z_m) = \sqrt{s_B^2 + s_r^2 + u^2(\delta) + \sum_{n=1}^N c_n^2 \times u^2(G_n)} \quad (7.4)$$

There are two situations where these simplifications may lead to an underestimation of MU:

- Case 1. When a reference material certified or external reference material (CRM or ERM) is used. The standard uncertainty of the reference value $u(X)$ is provided by the manufacturer with the reference material when it is a CRM or by the proficiency testing organizer when it is an internal reference material (Section 7.4). This uncertainty is equivalent to the definitional uncertainty defined in the International Vocabulary of Metrology (VIM), $u(X) = u(Z_d)$.
- Case 2. When the standard addition method (SAM) is used. In this case, it is then mandatory to include in the measurement model (6.7) the consequence of the spiking process, which means keeping the quantity $\sum_{n=1}^N c_n^2 \times u^2(G_n)$ and estimating it.

But in most cases, after simplification, it becomes:

Intermediate precision variance (reminder)

$$s_{IP}^2 = s_B^2 + s_r^2 \quad (3.1)$$

Standard variance of Z_m

$$\begin{aligned} u^2(Z_m) &\approx s_B^2 + s_r^2 + u^2(\delta) \\ u^2(Z_m) &\approx s_{IP}^2 + u^2(\delta) \end{aligned} \quad (7.5)$$

Standard uncertainty of Z_m

$$u(Z_m) \approx \sqrt{s_{IP}^2 + u^2(\delta)} \quad (7.6)$$

To demonstrate that, in the most general case, the data collected for an accuracy profile can also be used to estimate the analytical part of MU, it is satisfactory to show that formula (7.5) is equivalent to formula (5.7), i.e. that the variance s_{TI}^2 used to define the β -ETI around Z is equivalent to the standard variance $u^2(Z_m)$. The quantities already presented in Section 5.3.1, relative to the β -ETI are used.

In formula (7.5), the variance of intermediate precision s_{IP}^2 is an estimate of the dispersion of the measurements under the influence of the “series effect” that integrates the major sources of the MU, as illustrated in Figure 7.2. This parameter can be estimated by an ANOVA and applying the formulas of (3.5) and (3.24). To obtain the standard variance of the bias $u^2(\delta)$ it is necessary to go back to pose the model of the bias:

Bias

$$\delta = \bar{Z} - X \quad (7.7)$$

Then, comes quite naturally the standard variance of the bias while X is known without uncertainty:

Standard variance of the bias

$$u^2(\delta) = s_Z^2 + 0 = s_Z^2 \quad (7.8)$$

The standard variance of the bias is equal to the variance of the grand mean, whose formula corresponds to the Eq. (3.21).

Variance of the grand mean

$$s_Z^2 = s_{IP}^2 \times \left(\frac{1}{IJQ} \right) \quad (3.21)$$

Standard variance of the bias

$$u^2(\delta) = s_Z^2 = s_{IP}^2 \times \left(\frac{1}{IJQ} \right) \quad (7.9)$$

with:

$$Q = \frac{A+1}{J \times A+1} \quad A = \frac{s_B^2}{s_r^2}$$

Finally:

Standard variance of Z_m

$$\begin{aligned} u^2(Z_m) &\approx s_{IP}^2 + s_{IP}^2 \times \left(\frac{1}{IJQ} \right) \\ u^2(Z_m) &\approx s_{IP}^2 \times \left(1 + \frac{1}{IJQ} \right) \end{aligned} \quad (7.10)$$

Standard uncertainty of Z_m

$$u(Z_m) \approx s_{IP} \times \sqrt{1 + \frac{1}{IJQ}}$$

Standard deviation of β -ETI

$$s_{TI} = s_{IP} \sqrt{1 + \frac{1}{IJQ}} \quad (5.7)$$

Standard uncertainty of Z_m

$$u(Z_m) \approx s_{TI}$$

The conclusion is clear, and the standard uncertainty, calculated for the generic measurement model, is approximately equal to the standard deviation used to calculate the β -ETI. This result can be explained by the fact that most of the sources of variation have been grouped into the “series effect.” Furthermore, to construct the accuracy profile and calculate β -ETIs, it is mandatory to perform measurements according to the design of the experiment, which aims to alter the series effect and estimate the routine variability of the method. Then, to obtain a coverage interval, the coverage factor is set, allowing to determine the expanded uncertainty $U(Z)$. Two solutions are available for the coverage factor:

- Use the k_{TI} which participates in the β -ETI and puts the value of $\beta\% = 95\%$, representing a 95% coverage probability. This is possible as the number of degrees of freedom is equal to N_E . This is the best solution and is recommended as the first choice, among others, by the GUM.